

Question	Answer	Marks
3(a)	$\omega^2 = 2g / L$	C1
	$T = 2\pi / \omega$	C1
	$\omega^2 = (2 \times 9.81) / 0.19$ $\omega = 10.2 \text{ (rad s}^{-1}\text{)}$ $T = 2\pi / 10.2$ $= 0.62 \text{ s}$	A1
3(b)(i)	e.g. viscosity of liquid/friction within the liquid/viscous drag/friction between walls of tube and liquid	B1
3(b)(ii)	(maximum) $\text{KE} = \frac{1}{2}mv_0^2$ and $v_0 = \omega x_0$ or energy = $\frac{1}{2}m\omega^2 x_0^2$	C1
	change = $\frac{1}{2} \times 18 \times 10^{-3} \times 103 \times [(2.0 \times 10^{-2})^2 - (0.95 \times 10^{-2})^2]$	C1
	= $2.9 \times 10^{-4} \text{ J}$	A1

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4(c)	pulses (of ultrasound from generator)	B1